

Akanksha Sachan

CONTACT

Joint Carnegie Mellon University-University of Pittsburgh PhD Program aks203@pitt.edu
Aging Institute & Dept. of Immunology, School of Medicine, UPitt [LinkedIn](#)
Machine learning for Genomics & Epigenetics [Twitter](#) | [Github](#) | [Google Scholar](#)

EDUCATION

University of Pittsburgh '24 – Present
PhD Candidate in Computational Biology
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University '22 – '24
PhD Student in Computational Biology
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay '18 – '22
BTech. in Chemical Engineering | GPA: 8.06/10

AWARDS AND HONORS

American Federation for Aging Research (AFAR) Graduate Scholarship '25
Best Poster Award at the Aging Institute Research Day, UPitt '24
1st Place, Student A100 track NeurIPS LLM Efficiency Challenge | team of 5 '23
\$500 Track Prize (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 '22
Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay '22
Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship '21
Awarded a full scholarship (\$5,000) for iSURE undergraduate research program by the University of Notre Dame '20
Top 0.2% (Rank 3188 out of **1.1M+** candidates) in JEE (Main) '18
Top 1.5% (Rank 2517 out of **160k+** candidates) in JEE (Advanced) '18

INVITED TALKS

"FIREFate: Functional and Interpretable Regulatory Encodings of cell Fate" | *paper-in-prep*
Keystone Symposia: Artificial Intelligence in Molecular Biology Sep '25
International Society for Computational Biology (ISCB)/(ISMB) Jul '25, '26
American Association of Immunologists (AAI) Annual Meeting 2026 Apr '26
Center of Lung Aging and Regeneration, School of Medicine, UPitt Nov '25
30th Annual School of Medicine Graduate Student Research Symposium, UPitt Oct '25
Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat Aug '25
"Metabolic rewiring of skeletal muscle driven by loss of DNA-repair" | *paper-in-prep*
Center for Systems Immunology Annual Retreat, UPitt Nov '25
23rd Annual Department of Immunology Scientific Retreat, UPitt Sep '25
Aging Institute Research Day, UPitt Nov '24
"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"
50th European Calcified Tissue Society (ECTS) Congress Apr '23

ATHLETIC ACCOLADES

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay '19
3rd Position | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU '23
Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee '19
Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins '24
Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins '23

LEADERSHIP & SOCIETAL CONTRIBUTIONS

Vice President Aug '25 – Present

Allegheny Science Policy and Governance (ASPG)

Leading a recognized chapter of the **National Science Policy Network**; bridged scientists and PA legislators on **AI/data-center and aging policy** at the State Capitol, Harrisburg.

Teaching Assistant Fall '23 – Spring '24

02-760: Lab Methods for Computational Biologists, SCS, CMU

Taught a lecture on **genomics technologies** and gene regulation to a class of ~20 students.

Coordinator Aug '23 – Aug '25

Technights, Carnegie Mellon University, Robotics Institute

Led a **computer-science outreach** program teaching middle-school girls data analysis, foundation models, and deep learning.

Team Lead Mar '24

4D Nucleome Consortium Hackathon, Seattle, Washington

Led a team of 5 to build integrative pipelines for **single-cell HiC** analysis from cell-atlas-scale datasets; delivered a 30-minute presentation to consortium researchers.

Treasurer • Senator Aug '23 – Aug '25

Graduate Student Association (CPCB GSA)

Managed a **\$5,121** budget across CMU and UPitt events; organized the 2024 open-house for admitted CPCB students.

Editorial Board Member • Design Convenor '19 – '21

Insight, the student newsletter of IIT Bombay

Sketched & illustrated **1 Flagship & 1 Special Edition** magazine cover and a merch t-shirt; contributed editorial oversight and article curation.

Convenor, Girls Swim Team '19 – '20

Aquatics, Indian Institute of Technology Bombay

Led logistics, training coordination, and community-building events for the institute girls' team.

CONSORTIUMS & WORKSHOPS

SenNet Consortium Annual Meeting <i>Cellular Senescence Network, NIH, Washington, DC</i> Attended talks on cellular senescence atlasing and organ-specific senescence signatures.	Sep '25
Grace Hopper Celebration <i>AnitaB.org, Philadelphia</i> Talks and brainstorming sessions with tech-industry leaders.	Oct '24
Computational Genomics Summer Institute <i>University of California, Los Angeles</i> Delivered the lightning talk "Connecting 3D genomic organization to gene expression."	Jul '23
Grad Cohort for Women <i>Computing Research Association (CRA-WP), San Francisco</i> Mentoring sessions with women in CS across industry and academia.	April '23

RESEARCH EXPERIENCE PRIOR TO THESIS

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture <i>Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU</i> Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous cell lines; implemented hypergraph-based representation learning for integrative analysis of 3D genome architecture in single-cell, multi-omic mouse-brain datasets.	'22 – '24
Translational Mechanobiology Lab, MBI, National University of Singapore <i>Summer Internship Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine</i> Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support robust clinical deep-learning deployment; reviewed domain-adaptation and distribution-shift methods for liver-fibrosis staging from annotated medical images.	'22
Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich <i>Summer Internship Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech Presentation</i> Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center ; validated against in-vivo data from 235 diabetic patients and identified osteoid production as a key parameter via sensitivity analysis .	'21

**KEY
COURSEWORK**

Multicellular Systems Engineering Lab, University of Notre Dame

'20

Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering | Poster

Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide signaling; evaluated variance under gene-knockout in the Ca^{2+} signaling pathway using PCA features.

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

Predicting genomic-hub mediated TF-perturbation response via geometric deep learning

'23

10-708: Probabilistic Graphical Models | Code Repo

Augmented the GEARS GNN for Perturb-seq response prediction with a Hi-C contact graph (K562 **cancer cell-line** Micro-C, 20 kb bins mapped to gene TSSs), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs

'23

02-710: Computational Genomics | Code Repo

Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

Efficient Online Continual Semi-Supervised Learning

'22

10-701: Intro to ML

Improved semi-supervised OCL-PDS with active learning and optimized pseudo-labeling; outperformed the ER-FIFO method on the Amazon-WPDS dataset.